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Decision Support

Stochastic competitive entries and dynamic pricing

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ABSTRACT

How should firms price new products when they do not know the timing, nor the nature of the next competitive entry? To guide managers' pricing decisions in such contexts, we propose a dynamic pricing model with two types of randomly timed entry, i.e. imitative and innovative. The characterization of the equilibrium strategies reveals how optimal prices vary with the manager's knowledge about the timing of future competitive entries. We show that price skimming is not always optimal when entry dates are unknown to managers. Everything else equal, we demonstrate that the randomness of competitive entries make forward looking managers to choose constant prices, even though the characteristics of the market would have justified skimming the demand in the normal course. Moreover, we show that the constant pricing policy remains optimal even when the incumbent's optimal pricing strategy influences the probability of facing a competitive entry. Finally, we find that uncertainty does not necessarily hurt firms' profits.

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1. Introduction

Competitive entries permanently alter market structures. When new competitors leverage radically new technologies and innovate, incumbents' products become obsolete and stop generating sales (Bower & Christensen, 1995). For example, the emergence of the hydraulics technology in the mechanical excavator industry became lethal to incumbent firms. Innovative players, such as John Deere, Ford or Komatsu, incorporated this new technology into their products, which prompted the exit of most incumbents (Christensen, 1997, p. 64).

Alternatively, new entrants can emulate existing technologies and launch imitative products. In such cases, the new competition abruptly confronts incumbents with the daunting task of swiftly revamping marketing plans to prevent market-share losses. For example, Circuit City's surprised announcement of the launch of the DVIX format hurt sales of DVD players and forced the firms that were supporting the DVD format to suddenly rethink their sales forecasts and to change their marketing-mix (Besanko, Dranove, Shanley, & Schaefer, 2009, pp. 299–300).

In such contexts, the main challenge encountered by managers to optimize new products' marketing plans is to predict when competitive entries will take place (Geroski, 1999). However, as Moore, Oesch, and Zietsma (2007, p. 440) note: "potential entrants [base] their decision to enter primarily on evaluations of their own competence (or incompetence) and [pay] relatively little attention to the strength of the competition," which limits the use of economic theory to anticipate new entrants' behavior. Moreover, new entrants often come from outside of the focal industry, which also explains why managers regularly fail to identify competitive threats (Christensen, 1997). Finally, new technologies emerge as the random success of investments in research and development, making even harder to forecast entry dates (e.g. Kamien & Schwartz, 1983; Pennings & Sereno, 2011).

As a result, stochastic competitive entries prevent managers from smoothly steering products over the lifecycle. These random shocks complexify the planning of marketing decisions and exacerbate the risk of misallocating sales over time (Eliashberg & Jeuland, 1986). Indeed, if the incumbent envisions an early imitative entrant, she will likely choose a low price to ensure that her product achieves high penetration before the appearance of the imitative product. But if the actual entry occurs later than expected, she has sacrificed profit in anticipation of the impending competition. On the contrary, if the manager foresees a late competitive entry, she can choose a high price to take advantage of her expected longer monopoly position. If the actual entry happens earlier than initially envisioned, she will again lose profit by having left more unserved customers at the time of entry than initially expected. Accordingly, the incumbent over-allocates sales to the monopoly regime in the first case, whereas she over-allocates sales to the competitive regime in the second case.

Thus, stochastic competitive entries shake markets at random times and give birth to structural uncertainties (Basar & Haurie, 1984). Since Kamien and Schwartz (1971) and Gaskins (1971), numerous studies have analyzed the design of dynamic pricing strategies when managers face threats of homogeneous competitive entries (see Chatterjee, 2009 for a recent review). Kamien

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and Schwartz (1971) for instance show that a price skimming strategy is optimal when the manager faces the risk of encountering an imitative entry and where the hazard rate of entry increases with the incumbent's price. Bourguignon and Sethi (1981) extend the Kamien and Schwartz's framework to more general hazard rates whereas Bensoussan and Sethi (2007) provides a fully stochastic treatment of the KS model. While focusing on the impact of the pre-entry pricing strategy on the probability of entry, Kamien and Schwartz (1971) and Bourguignon and Sethi (1981) did not investigate how the pre-entry pricing strategy will affect post-entry sales, and consequently future cash-flows. Gaskins (1971), Deshmukh and Chikte (1976) and Fruchter and Messinger (2003) partially address this question. In these models, the pre-entry pricing strategies influence future profit due to the impact of the market price on the number of competitors, i.e. the rate of entry increases with the equilibrium price, while prices decrease with the number of players in the market. Hence, existing studies provide useful managerial guidelines about pricing strategies when stochastic competitive entries are identical in nature and when demand is never exhausted.

However in many situations this is not the case. Entries can differ and saturation effects often characterize demands for new products. These two issues are important because current market penetration will negatively impact future demand and because the inevitable innovative entry will put a sudden halt to the viability of the current technology. Consequently, not knowing when competitors will enter the market makes managers' pricing decisions more complex as the misallocation of sales today will have long-term impacts on future cash-flows through the saturation effects.

Therefore, how should managers price new products when they do not know the timing, nor the nature of the next competitive entry? Does price skimming remain optimal in such situations? Finally, how does the random nature of future entries affect profits?

To provide new managerial guidelines about pricing decisions under stochastic competitive entries, we propose a dynamic model that accounts for structural uncertainty. This new framework contributes methodologically to the differential game literature in marketing in general (Jørgensen & Zaccour, 2004; DeGiovanni, 2011; Erickson, 2012; Huang, Leng, & Liang, 2012) and to the dynamic pricing literature in particular (e.g. Chatterjee, 2009, Chenavaz, 2012), by incorporating the possibility of random regime changes set in motion "by finite state jump processes" (Basar & Haurie, 1984, p. 163). It preserves the negative effects of penetration on demand, accounts for the link between the pre-entry strategy and the post-entry sales and contrary to Chen and Jain (1992), allows for shocks that change the dynamics rather than the level of sales. The characterization of the equilibrium strategies reveals how managers' knowledge about the timing of future competitive entries shapes the optimal price trajectories. We obtain new results that challenge conventional wisdom in the dynamic pricing literature.

We first show that contrary to most new product pricing models, implementing a price skimming strategy when entry dates are unknown hurts firms' long-term profits. Everything else equal, we demonstrate that the randomness of competitive entries make forward looking managers to choose constant prices. This is true even though the characteristics of the market would have justified skimming the demand in the normal course.

This result occurs because time fails to provide any strategic information about the duration of each regime. The terminal date of the planning horizon becomes uncertain which prevents managers to work backward in the determination of the optimal price trajectories. The transversality conditions change, i.e. instead of solving two-point-boundary-value-problems, the stochastic entries confront managers with initial value problems.

Using these findings, we then numerically compare firms' long-term profits for scenarios in which managers' information sets vary with respect to the dates of two competitive entries. We find that firms' value functions can be higher under uncertainty than under certainty. This insight suggests that not knowing when competitors will enter the market is not necessarily detrimental to long-term profits. Why? Because, by implementing constant pricing strategies, managers allow firms to progressively capture higher unit margins compared to what pricing strategies under certainty would yield. At the same time, the randomness of competitive entries makes firms prefer early cash-flows over future ones. Hence, firms profit from uncertainty because the constant prices empower managers to harvest higher cash-flows in early time periods (at the expense of lower cash-flows in future periods), when they are more "valued".

The rest of the paper evolves as follows. Section 2 introduces the model. Section 3 characterizes the optimal policies under uncertainty. Section 4 extends the model to allow different specifications for the imitative entry intensity, as well as for the salvage values. Finally, Section 5 concludes.

2. Model

2.1. Time and uncertainty

We consider a product manager who launches a new product at $t = 0$. The product's lifetime, $[0; T]$, is finite but random since the manager does not know with certainty the date at which a new technology will replace the current one, i.e. T. Over this uncertain lifetime, the incumbent also faces the risk of encountering an imitative entry at an unknown date τ. These contingencies yield a stochastic planning horizon since two cases could unfold. Either the imitative entry happens first, i.e. $\tau < T$, in which case the incumbent competes with the new entrant until the technological breakthrough takes place. Or, the technological breakthrough happens first, i.e. $\tau > T$, in which case the new technology forces the old one out of the market. These entries mark sudden shocks that transform the sales dynamics. For instance, when going from the monopolistic regime to the competitive regime, the incumbent's sales will instantly vary with both her own price and the new entrant's. Conversely, the occurrence of an innovative entry will suddenly halt sales of existing products.

Therefore, stochastic entries delineate random jumps that are characterized by their intensity. Let $\{\Gamma(t)\colon t \geqslant 0\}$ be the the jump process governing the entry of the imitative product, i.e. $\Gamma(t) = 0$ when no entry has occurred by t, and $\Gamma(t) = 1$ otherwise. Similarly, let $\{\Omega_0(t)\colon t \geqslant 0\}$ be the jump process governing the entry of the innovative product if $t < T < \tau$ and $\{\Omega_1(t)\colon t \geqslant \tau\}$ be the jump process governing the entry of the innovative product if $\tau < t < T$. Finally, let the corresponding jump intensities be $0 < \chi < 1$, $0 < \omega_0 < 1$ and $0 < \omega_1 < 1$, respectively. Thus, these stochastic processes are defined by

$$\lim_{dt\to 0}\frac{P[\Gamma(t+dt)=1|\Gamma(t)=0;\ \Omega_0(t)=0]}{dt}=\chi,\quad \lim_{dt\to 0}\frac{P[\Gamma(t+dt)=0|\Gamma(t)=1]}{dt}=0, \tag{1}$$

$$\lim_{dt\to 0}\frac{P[\Omega_0(t+dt)=1|\Omega_0(t)=0;\ \Gamma(t)=0]}{dt}=\omega_0,\quad \lim_{dt\to 0}\frac{P[\Omega_0(t+dt)=0|\Omega_0(t)=1]}{dt}=0, \tag{2}$$

$$\lim_{dt\to 0}\frac{P[\Omega_1(t+dt)=1|\Omega_1(t)=0;\ \Gamma(t)=1]}{dt}=\omega_1,\quad \lim_{dt\to 0}\frac{P[\Omega_1(t+dt)=0|\Omega_1(t)=1]}{dt}=0. \tag{3}$$

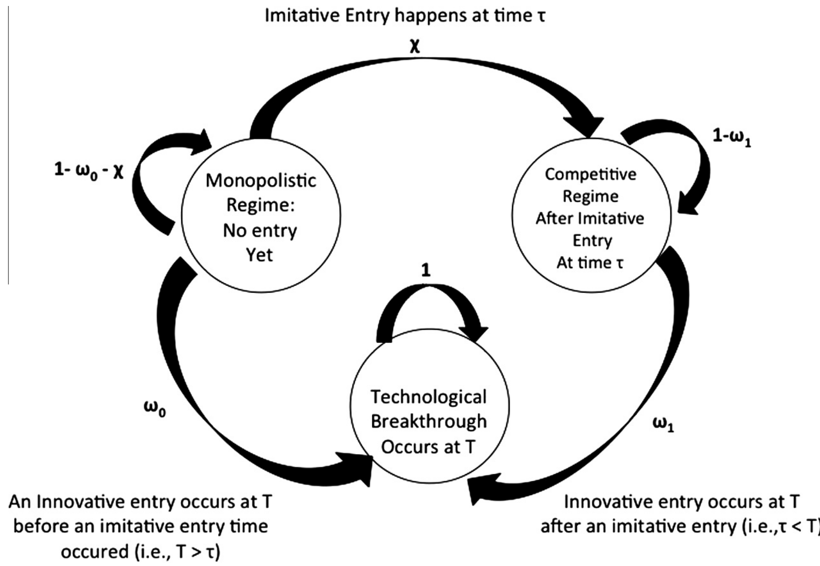

Fig. 1. Timeline of the game.

Hence, the timeline of the game is governed by finite state jump processes, which we summarize in Fig. 1.

We also note that these stochastic processes imply that the entry dates are exponentially distributed, which comports with the Schumpeterian theory of innovation (e.g. Aghion & Howitt, 1998), as well as with the industrial organization literature (e.g. Kamien & Schwartz, 1983).

2.2. Demands and profits

Following Eliashberg and Jeuland (1986), the sales dynamics in the monopoly regime is given by

$$\frac{dx_0(t)}{dt} = (m - x_0(t))(V_0 - \alpha_0 p_1(t)), \quad \text{with } x_0(0) = 0, \tag{4}$$

where m is the market potential, $x_0(t)$ is the incumbent's cumulative sales at time t, V_0 is the overall marketing advantage of the incumbent's product in the monopoly regime, α_0 is the price sensitivity and $p_1(t)$ is the price chosen by the incumbent. This motion equation implies that sales increase in proportion to the size of the remaining available market, which itself decreases with the product's penetration. When an imitative entry takes place, the demand dynamics for the incumbent and the new entrant, $i = 1, 2$ respectively, become

$$\frac{dx_i(t)}{dt} = (m - x_i(t) - x_{3-i}(t))(V_i - \alpha_i p_i(t) + \gamma(p_{3-i}(t) - p_i(t))), \tag{5}$$
$$\text{with } x_1(\tau^+) = x_1(\tau^-) \quad \text{and} \quad x_2(\tau^+) = 0,$$

with $p_i(t)$ the price of player i, γ the cross-price sensitivity, τ^- is the time right before the entry and τ^+ right after it. Dockner and Gaunersdorfer (1996) and Dockner and Fruchter (2004) rewrite the dynamics of sales by the dynamics of the pool of unserved customers, $z(t) = m - x_1(t) - x_2(t)$, such that the dynamics (4) and (5) become respectively

$$\frac{dz(t)}{dt} = -z(t)(V_0 - \alpha_0 p_1(t)), \quad \text{in the monopolistic regime,} \tag{6}$$

and

$$\frac{dz(t)}{dt} = -z(t)(V_1 + V_2 - \alpha_1 p_1(t) - \alpha_2 p_2(t)), \quad \text{in the competitive regime,} \tag{7}$$

with $z(0) = m$ and $z(\tau^+) = z(\tau^-)$. The introduction of this new state variable enlightens us that managing a new product compares to managing a non-renewable resource, i.e. the pool of unserved customers. Firms dynamically manage this resource through their prices. They can "consume" the resource faster by charging low prices; or they can conserve it by charging high prices. Managers seek to find the optimal allocation of sales across regimes by choosing pricing strategies that maximize their expected long-term profits. Specifically, the incumbent's profit function equals

$$W_0(t,z) = \max_{p_1} \mathbb{E}_{\Gamma,\Omega_0}\left\{\int_t^{\min\{\tau,T\}} D_0(s)(p_1(s)-c_0)ds + W_1(\tau,z(\tau))\cdot \mathbf{1}_{[\tau<T]} + B_1(T,z(T))\cdot \mathbf{1}_{[\tau>T]}\right\}, \tag{8}$$

subject to (6), with $\mathbf{1}_{[\tau<T]} = 1$ and $\mathbf{1}_{[\tau>T]} = 0$ if $\tau < T$ and *vice versa*. $D_0(s) = z(s)(V_0 - \alpha_0 p_1(s))$ gives the demand at time s and c_0 is the marginal cost of the incumbent in the monopolistic regime. Moreover, if the innovative entry happens first, the incumbent receives the salvage value $B_1(T, z(T))$, whereas if the imitative entry happens first ($\tau < T$), the monopolist obtains the long-term profit $W_1(\tau, z(\tau))$, which is the long-term profit associated with the competitive regime (i.e. $\tau < t < T$), defined as

$$W_i(t,z(t)) = \max \mathbb{E}_{\Omega_1}\left\{\int_t^T D_i(s)(p_i(s) - c_i)ds + B_i(T,z(T))\right\}, \tag{9}$$

for $i = 1$, subject to (7) with $D_i(s) = z(s)(V_i - \alpha_i p_i(s) + \gamma(p_{3-i}(s) - p_i(s)))$. Together, (6)–(9) define a stochastic differential game.[1]

To understand how the optimal strategies differ depending on which of τ or T is unknown, we analyze four scenarios. First, we replicate the findings of Eliashberg and Jeuland (1986) by allowing both dates to be known and ensuring that the imitative entry happens first. We solve the dynamic game using dynamic programming in continuous time. Second, we investigate the situation where both dates are unknown. Finally, we analyze how not knowing only one date impacts the optimal pricing strategies. Our analyses reveal that each scenario yields different optimal pricing strategies as summarized in Fig. 2.

[1] We assume, without loss of generality, that $B_i(T, z(T)) = 0$.

3. Dynamic pricing under certainty

In the deterministic case, the dates of both the imitative and innovative entries are known quantities and the monopolist chooses her pricing strategy, so as to maximize

$$U_0(t,z) = \max_{p_1} \int_t^{\tau} D_0(s)(p_1(s) - c_0)ds + \int_{\tau}^{T} D_1(s)(p_1(s) - c_1)ds, \tag{10}$$

subject to (6) and $D_0(s) = z(s)(V_0 - \alpha_0 p_1(s))$ for $s < \tau$, and (7), $D_1(s) = z(s)(V_1 - \alpha_1 p_1(s)) + \gamma(p_2(s) - p_1(s))$ for $\tau \leqslant s \leqslant T$. At the same time, the new entrant will choose his optimal pricing strategy, i.e. p_2, to maximize

$$U_2(t,z) = \max_{p_2} \int_t^{T} D_2(s)(p_2(s) - c_2)ds. \tag{11}$$

Together (6), (7), (10) and (11), describe the deterministic differential game analyzed by Eliashberg and Jeuland (1986). We characterize the corresponding Nash equilibrium by defining the current value Hamilton–Jacobi–Bellman (HJB thereafter) equation of the incumbent for $t < \tau$ as

$$-\frac{\partial U_0}{\partial t} = \max_{p_1} \left\{ D_0(p_1 - c_0) - z\frac{\partial U_0}{\partial z}(V_0 - \alpha_0 p_1) \right\}, \tag{12}$$

with the boundary condition $U_0(\tau, z(\tau)) = U_1(\tau, z(\tau))$. Similarly, for $\tau \leqslant t \leqslant T$, the HJB equation for $i = \{1, 2\}$ reads

$$-\frac{\partial U_i}{\partial t} = \max_{p_i} \left\{ D_i(p_i - c_i) - z\frac{\partial U_i}{\partial z}(V_1 + V_2 - \alpha_1 p_1 - \alpha_2 p_2) \right\}, \tag{13}$$

with the boundary condition $U_i(T, z) = B_i(T, z) = 0$. Solving for the optimal pricing strategies and value functions, we obtain the following result

Proposition 1. *Firms use price skimming when both dates are known and $\tau < T$. The incumbent's long-term profit is $U_0(t, z) = G(t)z$, whereas for $t > \tau$, $U_i(t, z) = H(t)z$.*

Under certainty, the realization of time provides strategic information to managers as it informs them about the exact duration of each regime, or said differently about how much time remains before witnessing a competitive entry. Forward-looking managers use it in finding the optimal introductory prices, rates of decline and curvatures of the price trajectories.

Time's strategic information is encapsulated in the partial derivatives of the value functions with respect to time, i.e. $\frac{\partial U_0}{\partial t}$ and $\frac{\partial U_i}{\partial t}$ in (12) and (13), respectively. At the equilibrium, the optimal monopolistic price decline is given by $\frac{\partial p_1}{\partial t} = \frac{1}{2z}\frac{\partial U_0}{\partial t} < 0$ for $t < \tau < T$, whereas competitive prices decline at rates equal to $\frac{\partial p_i}{\partial t} = \frac{\alpha_i}{(\gamma+\alpha_i)z}\frac{\partial U_i}{\partial t}$. Working backward from $t = T$ to $t = 0$, managers can then find the optimal introductory prices using the boundary conditions at $t = \tau$ and $t = T$.

Since the differential game is of the linear-state variety, we know that the solutions obtained by the HJB equations and the maximum principle are identical and represent the two sides of the same coin (see Dockner, Jørgensen, Long, & Sorger, 2000, pp. 187–194). This implies that the comparative statics results found in Eliashberg and Jeuland (1986) remains the same and that forward looking managers choose price skimming strategies under certainty because time informs them about how long they have to wait before witnessing a competitive entry.

4. Dynamic pricing under structural uncertainty

4.1. Competitive pricing with a stochastic innovative entry

To find the optimal policies under uncertainty, we work backward and first characterize the optimal strategies and value functions after the occurrence of an imitative entry. In this scenario, both the incumbent and the new entrant compete for the remaining available market until the innovative entry occurs. They choose the pricing strategies to maximize

$$W_i(t,z) = \max_{p_i} \mathbb{E}_{\Omega_1}\left\{ \int_t^{T-\tau} D_i(s)(p_i(s) - c_i)ds \right\}, \tag{14}$$

for $\tau < t < T$, where T is the only random variable since τ is now known. The program (14) can be recast as a deterministic dynamic problem. Over the interval $(t, t + dt)$, there is a given probability that the technological breakthrough will occur. Since the hazard rate of the innovative entry is defined by ω_1, the distribution function of T

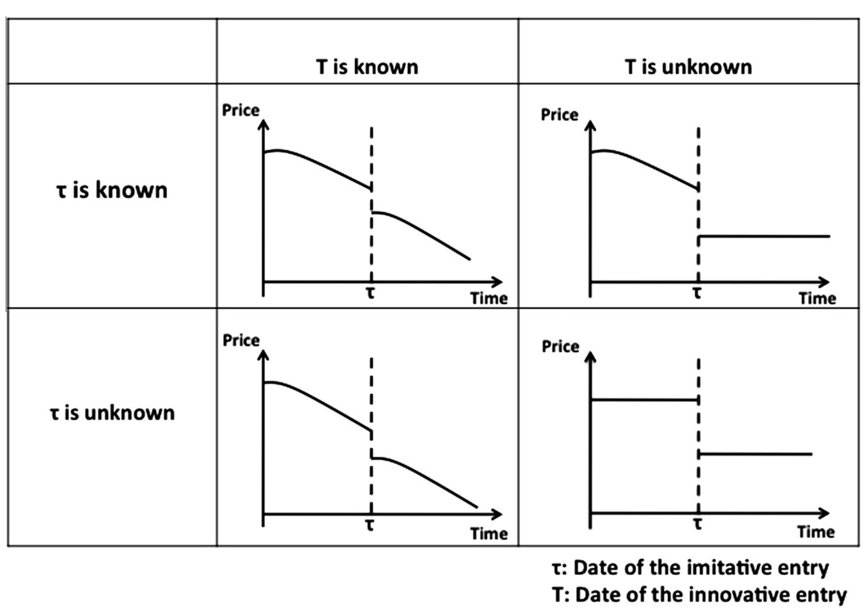

Fig. 2. Optimal price trajectories under uncertainty.

is $f_1(t) = 1 - e^{-\omega_1 t}$. Therefore, if no technological breakthrough took place before $\tau < t < T$, firms' expected long-term profits are

$$W_i(t,z) = \max_{p_i} \int_0^\infty \omega_1 e^{-\omega_1 s} \left\{ \int_0^t D_i(p_i - c_i) d\eta \right\} ds, \tag{15}$$

where $\omega_1 e^{-\omega_1 t} = \omega_1(1 - f_1(t))$ is the probability of facing a technological breakthrough given that no innovative entry took place before t. Integrating (15) by parts, we obtain

$$W_i(z) = \max_{p_i > 0} \int_0^\infty e^{-\omega_1 s} \{D_i(p_i - c_i)\} ds. \tag{16}$$

We define player i's HJB equation as

$$0 = \max_{p_i} \left\{ D_i(p_i - c_i) - z \frac{dW_i}{dz} (V_1 + V_2 - \alpha_1 p_1 - \alpha_2 p_2) - \omega_1 W_i(z) \right\}. \tag{17}$$

Solving for the optimal strategies and value functions in Appendix A, we obtain the following proposition.

Proposition 2. *Firms choose constant prices in the competitive regime when they do not know the date of the technological breakthrough.*

The randomness of the innovative entry prevents firms from inferring how close the technological breakthrough is, i.e. neither the incumbent nor the new entrant extract strategic information from time. Therefore, the term $\frac{\partial W_i}{\partial t}$ disappears from the firms' HJB equations (i.e. (17)) compared to the certainty case (i.e. (13)). In other words, time becomes irrelevant to firms' payoffs and forward-looking managers ignore it in the determination of the optimal policies. As a result, both competitors adopt constant prices in the competitive regime.

The comparative statics analyses first show that prices decrease in own-price sensitivities, i.e. α_i, and increase in firms' overall marketing advantages and marginal costs (V_i and c_i respectively). Second, prices decrease in ω_1, the intensity of Ω_1 and we also find that not knowing the date of the technological breakthrough, makes managers more present-oriented, as illustrated by $e^{-\omega_1 t}$ in (16). Hence, managers choose to increase the speed of diffusion of their products before an innovative entry forces them out of the market. Third, prices also decline as γ increases, which comports with findings of Eliashberg and Jeuland (1986, p. 30). The robustness of this result to the stochastic case implies that uncertainty does not prevent firms from strategically responding to existing competition. Even if they become more present-oriented and choose constant prices in the competitive regime, they still incorporate the other competitor's strategic behavior in their own decisions.

Finally, the optimal prices are higher under uncertainty than those under certainty as $t \to T$. Indeed, prices under certainty converge to the static prices as $t \to T$, whereas prices under uncertainty remain constant, and above the myopic prices. However, right after the competitive entry ($t \to \tau^+$), prices under uncertainty can be higher, or lower, than those under certainty, depending on the length of the competitive regime. On the one hand, if the duration of the competitive regime is long, i.e. $T = T_L$, firms have more time to skim the cream of the demand and choose to shift their prices upward. On the other hand, if the technology arrives early, implying a short duration for the competitive regime, i.e. $T = T_S$ (with $T_S < T_L$), firms will decrease their prices more rapidly in response to the imminent breakthrough. Under uncertainty firms cannot distinguish between the two cases, and so they optimally maintain constant prices. Therefore, two cases are possible; either prices under uncertainty are always higher than those under certainty (if $T = T_S$), or they are lower at first then higher later (if $T = T_L$), as depicted in Fig. 3.

Therefore, the randomness of the innovative entry induces managers to adopt pricing strategies such that they can extract more unit margins from sales as over time, compared to the certainty case.

4.2. Monopolistic dynamic pricing with two unknown dates

To characterize the incumbent's pricing strategy and value function in the monopolistic regime, we recall that the manager's dynamic problem is in that case to find the pricing strategy that maximizes her long-term profit, i.e.

$$W_0(t,z) = \max_{p_1} \mathbb{E}_{\Gamma,\Omega_0} \left\{ \int_t^{\min\{\tau,T\}} D_0(s)(p_1(s) - c_0) ds + W_1(\tau,z) \cdot \mathbf{1}_{[\tau<T]} + B_1(T,z) \cdot \mathbf{1}_{[\tau>T]} \right\}, \tag{18}$$

which means that her expected long-term profit depends on two random variables τ and T. We decompose (18) as $W_0(t, z(t)) = \mathrm{Max}_{p_1}\{I_\tau(t,z) + I_T(t,z)\}$, where $I_\tau(t, z)$ is the expected profit if the imitative happens first and $I_T(t, z)$ is the expected profit if the innovative entry happens first, defined respectively by

$$I_\tau(t,z) = \int_0^\infty \chi e^{-\chi s} e^{-\omega_0 s} \left\{ \int_0^t D_0(p_0 - c_0) d\eta + W_1(t,z) \right\} ds, \tag{19}$$

and

$$I_T(t,z) = \int_0^\infty \omega_0 e^{-\omega_0 s} e^{-\chi s} \left\{ \int_0^t D_0(p_0 - c_0) d\eta + B_1(t,z) \right\} ds, \tag{20}$$

where $\chi e^{-\chi t} e^{-\omega_0 t}$ refers to the probability that the imitative entry takes place at t taking into account that no entry took place before; whereas $\omega_0 e^{-\omega_0 t} e^{-\chi t}$ refers to the probability that the innovative entry takes place at t taking into account that no entry took place before. Integrating both $I_\tau(t, z)$ and $I_T(t, z)$ by parts, we obtain

$$W_0(z(0)) = \max_{p_1} \int_0^\infty e^{-(\chi+\omega_0)t} \{ D_0(t)(p_1(t) - c_0) + \chi W_1(t,z) + \omega_0 B_1(t,z) \} dt. \tag{21}$$

Letting $B_1(t, z) = 0$, the monopolist's Bellman equation becomes

$$0 = \max_{p_1>0} \left\{ D_0(p_1 - c_0) - z \frac{dW_0}{dz} (V_0 - \alpha_0 p_1) + \chi(W_1(z) - W_0(z)) - \omega_0 W_0(z) \right\}, \tag{22}$$

which comports with Theorem 8.3 in Dockner et al. (2000, p. 220). Solving for the optimal pricing strategy and value function, we obtain a new proposition.

Proposition 3. *The incumbent's pricing strategy is constant in the monopolistic regime when the dates of both competitive entries are unknown to the manager.*

Similar to the competitive case, the randomness of the jumps prevents the incumbent from inferring how close the competitive entries are. Since time does not provide any strategic information, the incumbent chooses a constant price. Therefore, under structural uncertainties, prices will be constant in both regimes. This insight is important since most new product pricing models establish that managers should skim the demand. In particular Eliashberg and Jeuland (1986), succeeded by Dockner and Jørgensen (1988) and Dockner and Fruchter (2004), show that the saturation effects, together with competition, suffice to warrant the implementation of skim pricing strategies under certainty. Propositions 2 and 3 show that is not the case when competitive entries are stochastic, i.e. when managers do not know when the next entry will take place, and therefore they should keep prices constant.

Does this insight imply that consumers do not adjust their adoption timing in response to firms' strategies? This question matters since one reason for skim pricing to be optimal is that consumers are heterogeneous, which makes them adopt new products

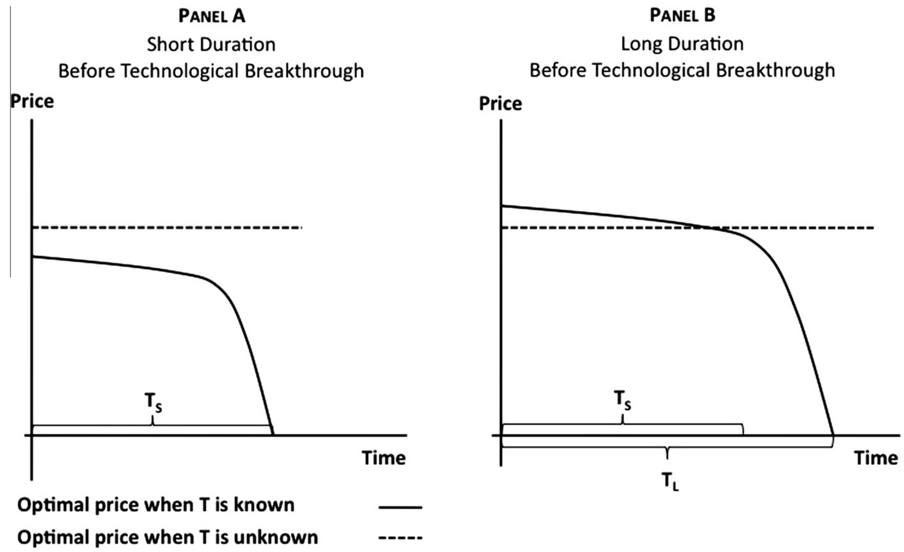

Fig. 3. Price under certainty vs. price under uncertainty.

at different times (e.g. Besanko & Winston, 1990). However, the micro-foundations of the sales dynamics used in our analysis imply that constant prices do not prevent the heterogeneity of consumers' preferences from driving the adoption process. Indeed, Chatterjee and Eliashberg (1990) show that even when new products' prices are constant, consumers acquire information at different rates. Each individual will then decide to make a purchase or wait based on his/her own behavioral characteristics and preferences. A customer will purchase the new product as soon as the information available in the market reaches a certain threshold value, which differs for each individual. Hence, despite the fact the optimal prices are constant under uncertainty, consumers will respond differently to market information and adjust their adoption timing in consequence, leading to different purchase times.

The comparative statics analyses of the monopolistic price with respect to the model's parameters reveal that p_1 decreases in ω_0, which is explained again by the fact that the incumbent becomes more present-oriented and consequently chooses to increase the penetration of her product before the technological breakthrough takes place. On the contrary, (21) reveals that χ has a dual role. On the one hand, χ increases the manager's time preference through $e^{-(\chi+\omega_0)t}$, which creates an incentive to decrease p_1, in order to realize more sales before the imitative entry takes place. On the other hand, when p_1 decreases, so does $z(t)$, which causes (21) to decrease as well through $\chi W_1(z)$. The incumbent then faces a tradeoff, where, similar to the certainty case, she has to find an optimal allocation of sales across regimes. As χ increases, she could increase or decrease p_0 depending on the link between her pre-entry strategy and her post-entry profit. Specifically, if the contribution of each additional sale to the long-term profit in the competitive regime is sufficiently high such that $\frac{\partial W_1}{\partial z} > \frac{(V_0+2\omega_0-c_0\alpha_0)-\sqrt{4\omega_0(V_0-c_0\alpha_0)-(V_0-c_0\alpha_0)^2+4\omega_0^2}}{2\alpha_0}$, it becomes optimal for the incumbent to shift sales from the pre-entry regime to the post-entry regime by increasing her price (see Appendix A for the sensitivity of p_1 with respect to χ). Otherwise, the incumbent should decrease her price as χ increases.

Finally, we note that the incumbent's pricing strategy varies with the characteristics of the market in the competitive regime, captured through $\frac{\partial W_1}{\partial z} = A_1$, i.e. $\frac{\partial p_0}{\partial A_1} > 0$. In particular, $\frac{\partial A_1}{\partial \gamma} < 0 \Rightarrow \frac{\partial p_0}{\partial \gamma} < 0$. As the monopolist envisions price competition to be fierce, she reduces her price in anticipation of new entry; she allocates more sales to the monopoly regime. In other words, the incumbent takes into account the likely post-entry market characteristics in her pre-entry pricing decisions.

4.3. Impacts of prices on entry probabilities

We assumed so far that the entry intensities are independent from the monopolist's pricing strategy. To investigate what would happen if this was not the case, we follow Kamien and Schwartz (1971) and relax this assumption to consider situations where the incumbent's price influences the entry hazard rates. We define $\tilde{\chi} = g(p_1)$ as the intensity of Γ, where $g(\cdot)$ is a function of p_1. Similarly, we define $\widetilde{\omega_0} = h(p_1)$ as the intensity of Ω_0.

Similar to Section 4.2, we integrate the monopolist's long-term profit by parts, which yields

$$L_0(z(0)) = \max_{p_1} \int_0^\infty e^{-(g(p_1)+h(p_1))t}\{D_0(t)(p_1(t)-c_0) + g(p_1(t))W_1(t,z)\}dt. \quad (23)$$

In this case, the monopolist's pricing strategy influences how she discounts future payoffs as revealed by $e^{-(g(p_1)+h(p_1))t}$ and the corresponding HJB equation equals

$$0 = \max_{p_1}\left\{D_0(p_1-c_0) - \frac{dL_0}{dz}z(V_0-\alpha_0 p_1) + g(p_1)(W_1(z)-L_0(z)) - h(p_1)L_0(z)\right\}. \quad (24)$$

Differentiating (24) with respect to price and conjecturing that the value function is proportional to z, such as $L_0(z) = \widetilde{A}_0 z$, we obtain

$$V_0 + \alpha_0(\widetilde{A}_0 + c_0 - 2p_1) + (A_1 - \widetilde{A}_0)\frac{\partial g}{\partial p_1} - \widetilde{A}_0\frac{\partial h}{\partial p_1} = 0. \quad (25)$$

The first-order condition (25) implies that the optimal pricing strategy remains constant over time for any functions $g(\cdot)$ and $h(\cdot)$, i.e. $\frac{dp_1}{dt} = 0$. Hence, even when the monopolist can influence the hazard rate of entry, she will charge a constant price. However, the optimal policy differs from the strategy in Section 4.2. First, the condition (25) implies that the incumbent's optimal pricing strategy is augmented by $(A_1 - \widetilde{A}_0)\frac{\partial g}{\partial p_1} - \widetilde{A}_0\frac{\partial h}{\partial p_1} = \left(\frac{dW_1}{dz} - \frac{dL_0}{dz}\right)\frac{\partial g}{\partial p_1} - \frac{dL_0}{dz}\frac{\partial h}{\partial p_1}$, which measures the marginal variation of the incumbent's long-term profit, engendered by switching from one regime to another. Second, the identification of $\frac{dL_0}{dz} = \widetilde{A}_0$ will yield a different quantity compared to what is obtained in the situation where $g(p_1) = \chi$ (see (36) in

Appendix A). Therefore, contrary to Kamien and Schwartz (1971), we find that accounting for the link between the pre-entry strategy and the post-entry profit makes the incumbent's optimal price remain constant over time, even though it will be different than when prices do not influence entry hazard rates.

4.4. Dynamic pricing with one known date

To further understand how stochastic competitive entries change firms' pricing decisions, we first characterize the optimal strategies and value functions when only the date of the innovative entry is known. Then, we look at the other case, where only the date of the imitative entry is known. The characterizations of the optimal pricing strategies in both cases lead to the following results.

Proposition 4. *Price skimming is optimal in both regimes when the date of the innovative entry is the only date to be known.*

Proposition 5. *Price skimming is optimal only in the monopolistic regime when the date of the imitative entry is known while the date of the technological breakthrough is unknown.*

Both propositions comport with our previous findings in the sense that when firms can extract strategic information from time, they implement price skimming strategies. In the first case, the information comes from the remaining time before the innovative entry takes places. In this scenario, the incumbent always knows how much time is left before a new technology appears. Therefore, she decreases her price over time. If an imitative entry takes place before the technological breakthrough happens, she switches to a new price that she had pro-actively pre-determined at the beginning of the planning horizon when envisioning the possibility of the imitative entry.

If she had ignored that such imitative entry could happen, she would still change the product's price at the time of entry but such that the size of the "jump" between the monopolistic price and the competitive one would be bigger than the optimal one. In the second case, the incumbent implements a price skimming strategy until the imitative entry takes place. But after this entry, both competitors choose constant prices because none of them knows when the innovative entry will happen.

Hence, Propositions 1–5 offer novel insights into the optimal dynamic pricing strategies under uncertainty. Price skimming is optimal only when time provides firms with strategic information concerning future competitive entries. Moreover, we learn that uncertainty partially mutes firms' competitive reactions by preventing them from continuously adjusting their prices over time. However, $\frac{\partial p_i}{\partial \gamma}$ are different from zero, which suggest that firms still behave strategically in response to existing competitors and potential entrants.

5. Impacts of uncertainty on profits

What is the impact of structural uncertainty on firms' profits? To investigate this question and provide decision-theoretic insights, we numerically compare the value functions of both the incumbent and new entrant under two scenarios, all dates are known (certainty) and none is known (full uncertainty).

Under certainty, the incumbent's ex-ante long-term profit is then defined by $U_0(0, z(0))$, while the new entrant's ex-ante long-term profit after entry by $U_2(\tau, z(\tau))$, as given in Appendix A. Together, these two quantities allow us to compute total industry profits under certainty, i.e. $U_0(0, z(0)) + U_2(\tau, z(\tau))$. We then numerically compute the incumbent's and new entrant's value function when both dates are unknown, i.e. $W_0(0, z(0))$ and $W_2(\tau, z(\tau))$, respectively. Similar to the certainty case, these quantities allow us to compute total industry profits under uncertainty, i.e. $W_0(0, z(0)) + W_2(\tau, z(\tau))$.

5.1. Impacts on the incumbent

We compute (ex-ante) value functions for different jump intensities and entry dates to quantify the impact of uncertainty on the incumbent's profit.[2] More precisely, assuming that $\omega_0 = \omega_1 = \omega$, we compute for each pair (χ, ω) the difference δ between the value function under certainty and the value function under uncertainty, i.e. $\delta = \widetilde{U_0} - W_0$ such that $\widetilde{U_0} = \int\int U_0(0, z(0))\lambda_1[\tau]\lambda_2[T]d\tau\, dT$, where $\lambda_1(\tau)$ and $\lambda_2(T)$ are the PDFs of τ and T with parameters χ and ω, respectively.[3] For the value function under uncertainty, we replaced the numerical values of χ and ω in $W_0(0, z(0))$ and report the results for δ in Fig. 4.[4]

We first note that in most cases, the value function under certainty is higher than the value function under uncertainty. However, we also observe that in some instances the incumbent is better off not knowing when the competitive entries will take place. This happens in particular when the manager expects the imitative entry to happen later rather than sooner (i.e. low χ). In other words, the intertemporal allocation of sales generated by the optimal pricing rule under uncertainty, empowers the incumbent to profit from not knowing when competitive entries will take. In relative terms, the steep behavior represents around 30% gains in the value function under uncertainty compared to value function under certainty.

The reason for the incumbent to be better off not knowing when the new entrant will arrive can be explained by the fact that the optimal pricing strategy under uncertainty allows the incumbent to capture early cash-flows that are larger than those obtained in the certainty case (see Fig. 2), even if it means that later cash-flows will be lower due to lower sales. Simultaneously, the mere fact that competitive entries are stochastic makes the incumbent more present oriented. Therefore these early (and larger) cash-flows carry more weight in the incumbent's value function. As a result, the value function under uncertainty becomes larger than the value function under certainty.

Could this insight hold true in the particular case where the expected entry dates in the stochastic model equal the (known) entry dates in the deterministic model?[5] In other words, is it possible to have $U_0(0, z(0)) < W_0(0, z(0))$ when $\mathbb{E}[\tau] = \tau$ and $\mathbb{E}[T] = T$? To answer this question, we recall that under uncertainty, τ and T are random variables that are exponentially distributed. This implies that $\mathbb{E}[\tau] = \tau \Rightarrow \chi = 1/\tau$ and $\mathbb{E}[T] = T \Rightarrow \omega_0 = \omega_1 = 1/T$, which we use to compute ex-ante value functions under both uncertainty and certainty. We find that depending on the value of parameters, it is possible to not only have $U_0(0, z(0)) > W_0(0, z(0))$, but also $U_0(0, z(0)) < W_0(0, z(0))$. For instance in Fig. 5, $U_0(0, z(0)) > W_0(0, z(0))$ is obtained with $V_0 = V_1 = V_2 = .1$, $c_0 = c_1 = c_2 = 500$, $\gamma = 0.0001$, $\alpha_0 = \alpha_1 = \alpha_2 = 0.00005$, $t = 0$ (Scenario 1). Alternatively, $U_0(0, z(0)) < W_0(0, z(0))$ is obtained by only changing the intercepts of the demand functions to $V_0 = V_1 = V_2 = .01$ (Scenario 2).

Therefore, Figs. 4 and 5 together suggest that structural uncertainty does not necessarily hurt the incumbent's long-term profit. What about the new entrant and the impacts of structural uncertainty on total industry profits?

[2] We thank an anonymous reviewer for suggesting this approach.

[3] We used the function NExpectation in Mathematica to numerically evaluate $\widetilde{U_0}$.

[4] The numerical simulations were performed with the following values: $m = 1$, $V_0 = V_1 = V_2 = .1$, $c_0 = c_1 = c_2 = 500$, $\gamma = 0.0001$, $\alpha_0 = \alpha_1 = \alpha_2 = 0.00005$.

[5] We thank an anonymous reviewer for this suggestion.

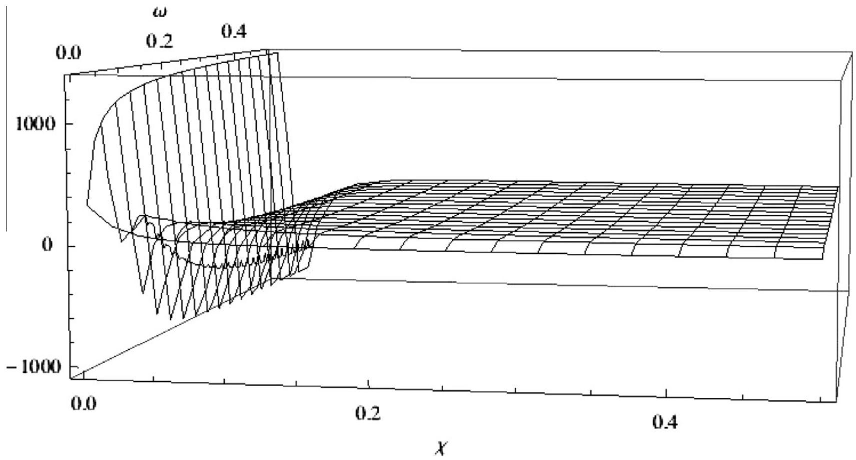

Fig. 4. Difference between ex-ante value functions under certainty and uncertainty.

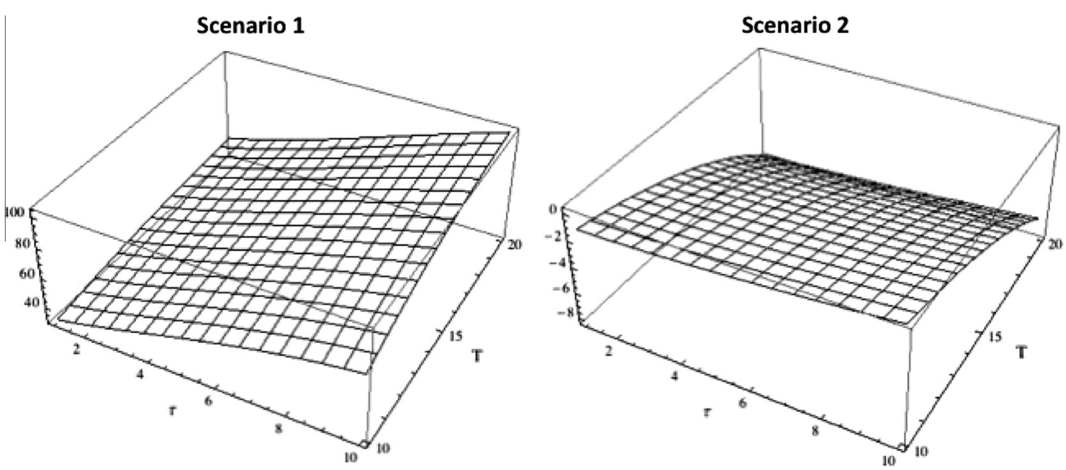

Fig. 5. Difference between ex-ante value functions under certainty and uncertainty when $\chi = 1/\tau$ and $\omega = 1/T$.

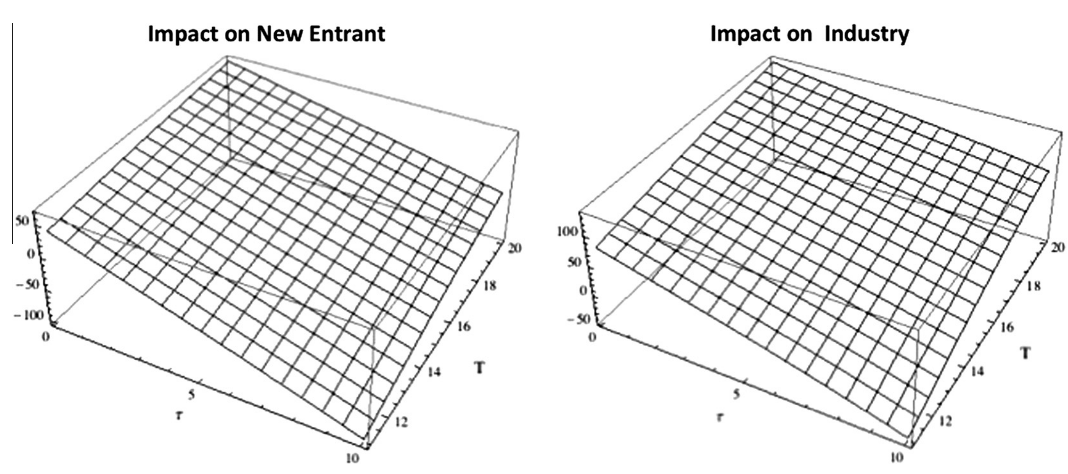

Fig. 6. Impact of uncertainty on the new entrant's and total industry profits $\chi = 1/\tau$ and $\omega = 1/T$.

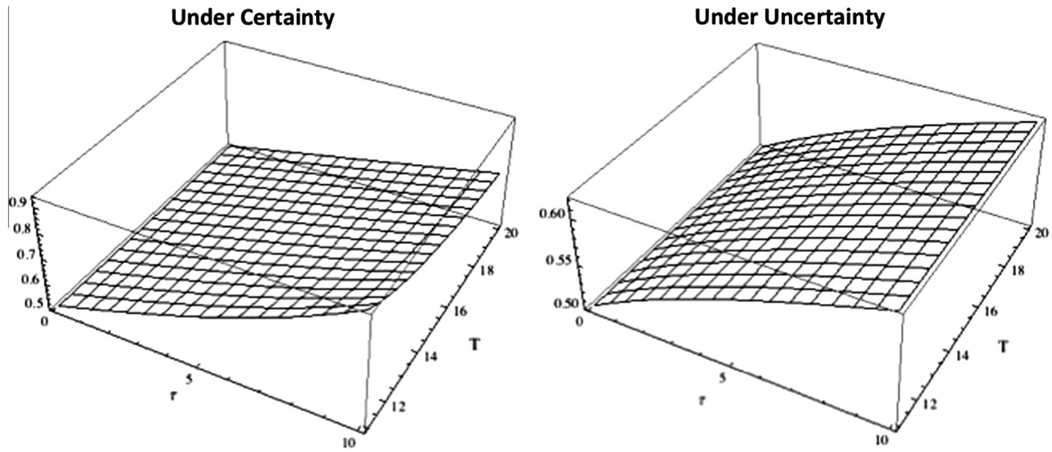

Fig. 7. Incumbent's share in total industry profits.

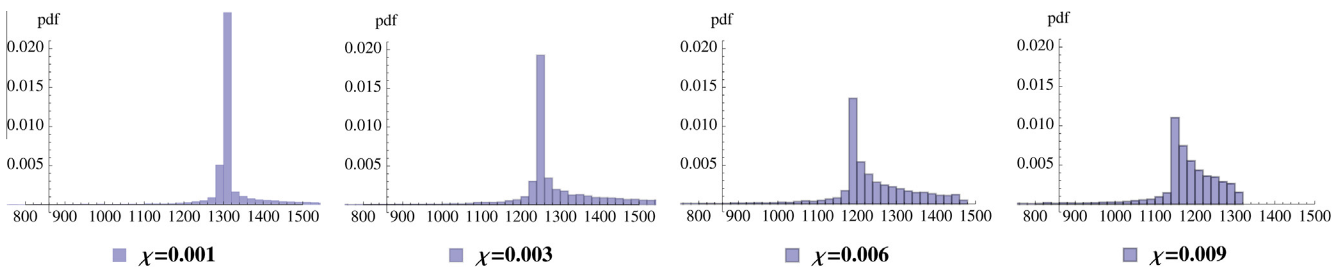

Fig. 8. Profit distribution for increasing values of χ.

Table 1
Properties of the distribution of uncertain ex-post profits.

	Mean	Standard deviation	Skewness	Profit under certainty
$\chi = 0.001$	1354.7	196.7	1.7	1424.1
$\chi = 0.003$	1302.2	170.3	−0.3	1297.8
$\chi = 0.006$	1223.6	147.2	−2.6	1150.5
$\chi = 0.009$	1167.3	129.3	−3.6	1038.9

5.2. Impacts on the industry

To address these questions, we report additional numerical results. These numerical simulations are performed using the same parameter values as in Scenario 1 in Fig. 5 above, as well as assuming that $\mathbb{E}[\tau] = \tau$ and $\mathbb{E}[T] = T$. We report on the left panel of Fig. 6 the difference, for the new entrant, between profit (value function) under certainty and profit under uncertainty. On the right panel of Fig. 6, we report the difference between total industry profits under certainty and uncertainty. We also report with Fig. 7 the share of the incumbent in the total industry profits under certainty and under uncertainty.

We first note that the new entrant's value function under uncertainty can be both lower or higher than his value function under certainty. This insight corroborates with what we found for the incumbent. In addition, we learn that the incumbent's captures the lion's share of total industry profits under both uncertainty and certainty (see Fig. 7). This is not surprising due to the competitive advantage offered by incumbency in this game.

Finally, to complete our numerical analyses, we sample 10,000 scenarios from the distributions of the entry dates with $V_0 = V_1 = V_2 = .1$, $c_0 = c_1 = c_2 = 500$, $\gamma = 0.0001$, $\alpha_0 = \alpha_1 = \alpha_2 = 0.00005$, $\omega = 0.001$ and varying χ from 0.001 to 0.009. We report the distribution of ex-post profits in Fig. 8 and provide additional information about the distribution of ex-post profits in Table 1.

Fig. 8 and Table 1 allow us to gain more insights about the impacts of uncertain entry dates on profit distribution. First, we learn that even ex-post, firms can also profit from not knowing in advance when competitors will enter the market. But, we also learn that the skewness of the distribution is increasing as the difference between the certain and uncertain profit increases. Collectively, these observations allow us to obtain a new proposition.

Proposition 6. *Firms can profit from not knowing the dates of competitive entries.*

6. Conclusion

Price skimming is a cornerstone recommendation of the dynamic pricing literature (e.g. Rao & Kartano, 2009; Robinson & Lakhani, 1975). It consists in monotonically decreasing the price of new products over time to maximize profits. We discovered that this strategy is perilous when managers do not know neither the date, nor the type (innovative vs. imitative) of the next competitive entry. In such cases, we find that firms should instead maintain constant prices. To obtain this novel insight, we proposed a dynamic model that captures how stochastic competitive entries create structural uncertainties that permanently alter market structures. We find that the constant pricing strategy holds true even when the hazard rates of entry are functions of the incumbent's price. This insight implies that even if incumbent firms know that their prices influence the behaviors of potential competitors, "keeping still" remains optimal and that it is possible that firms' value functions under uncertainty exceed those under certainty, depending on the model's parameters.

Moreover, we find that competitive entries induce price trajectories to be piecewise-continuous. This theoretical insight comports with Hernandez-Mireless, Fok, and Franses (2012, p. 24) who not only document the importance of such phenomena in technology markets, but who also empirically show that "*the most likely trigger of price landings* (...) *is competitive entry*." Our results provide a theoretical explanation for such phenomena and provide a way to link the magnitude of these sudden price drops to observable market factors *as well as* to unobservable managerial traits, i.e. managers' expectations with respect to the dates of these competitive entries.

How can firms then optimally manage these price landings, especially when they do not know when they will have to implement them? The answer we provide is to be pro-active instead of being reactive at the time of entry. Incumbents should prepare themselves for the risk of competitive entry by optimally choosing pricing strategies that explicitly account for *random* regime switches. In other words, our approach provides managers with a tool to implement pricing strategies that account for disruptive events. Moreover, since our findings show how prices encapsulate managers' expectations about future competitive entries, marketing research firms that specialize in the marketing of new products and new technologies, such as Ipsos Vantis or Forrester Research, can leverage our theoretical results to empirically quantify these expectations to help their clients reach their business goals. Finally, the last managerial recommendation derived from our results is to advise managers not to implement price skimming strategies when they do not know when competitors will enter.

To conclude, this research can be extended in several directions. First, considering marketing decisions other than price would be interesting to understand how firms should manage the marketing mix of new products under uncertainty (e.g. Kamrad, Lele, Siddique, & Thomas, 2005). Second, considering sales dynamics exhibiting diffusion effects would be interesting to further understand optimal pricing decisions under structural uncertainty. Finally, we assumed that the technological breakthrough was independent from firms' actions. Relaxing this assumption would provide a new perspective on how firms should invest in research and development when they envision possible competitive entries (Chevalier-Roignant, Flath, Huchzermeier, & Trigeorgis, 2011; Lambrecht & Perraudin, 2003).

Appendix A

A.1. Proof Proposition 1

The proof proceeds backwards, starting with the identification of the value function after the imitative entry took place at $t = \tau$. The HJB equation for player i reads

$$-\frac{\partial U_i(t,z)}{\partial t} = \max_{p_i}\left\{D_i(p_i - c_i) - z\frac{\partial U_i}{\partial z}(V_1 + V_2 - \alpha_1 p_1 - \alpha_2 p_2)\right\}, \tag{26}$$

with the boundary conditions $U_i(z, T) = 0$. Assuming firms are symmetric, the first order conditions is

$$p_C = p_1 = p_2 = \frac{V_1 + \alpha_1 \partial U_1/\partial z + c_1(\gamma + \alpha_1)}{\gamma + 2\alpha_1}. \tag{27}$$

The structure of (26) reveals that the game played between the two firms in the competitive regime is of the linear state variety (see Dockner et al., 2000, pp. 187–194). Hence, we conjecture that the structure of the value function is linear in the state, i.e. z, such as $U_1(t, z) = U_2(t, z) = H(t)z$, where $H(t)$ is a coefficient that has to be identified. We do so following the Lagrange's method of undetermined coefficients. Specifically, after inserting (27) in (26), the coefficient of the value function is characterized by the solution of

$$\dot{H}(t) = q_1 H(t)^2 + q_2 H(t) + q_3, \tag{28}$$

with $H(T) = B_i(T) = 0, q_1 = -\frac{\alpha_1^2(2\gamma+3\alpha_1)}{(\gamma+2\alpha_1)^2}, q_2 = \frac{(V_1-c_1\alpha_1)(2\gamma^2+5\alpha_1\gamma+4\alpha_1^2)}{(\gamma+2\alpha_1)^2}$ and $q_3 = -\frac{(\gamma+\alpha_1)(V_1-c_1\alpha_1)^2}{(\gamma+2\alpha_1)^2}$. The solution of (28) is

$$H(t) = -\frac{\sqrt{q_2^2 - 4q_1q_3}\tanh\left(\tanh^{-1}\left(-\frac{q_2}{\sqrt{q_2^2-4q_1q_3}}\right) + 0.5\sqrt{q_2^2 - 4q_1q_3}(T-t)\right) + q_2}{2q_1}. \tag{29}$$

Note that differentiating (29) with respect to time yields

$$\frac{\partial H(t)}{\partial t} = \frac{q_3(q_2^2 - 4q_1q_3)}{\left((2q_1\psi + q_2)\sinh\left(\frac{1}{2}\sqrt{q_2^2-4q_1q_3}(t-T)\right) - \sqrt{q_2^2-4q_1q_3}\cosh\left(\frac{1}{2}\sqrt{q_2^2-4q_1q_3}(t-T)\right)\right)^2} < 0.$$

Hence, in the competitive regime, prices go down since $\frac{\partial p_C}{\partial t} = \frac{\alpha_1}{\gamma+2\alpha_1}\frac{\partial H(t)}{\partial t} < 0$. The post-entry long-term profit, i.e. $U_1(t, z)$, is included in the characterization of the incumbent's pre-entry value function as follows

$$U_0(0, z(0)) = \max_{p_1}\int_0^\tau D_0(t)(p_1(t) - c_0)dt + U_1(\tau, z).$$

Consequently, the HJB equation of the incumbent in the monopolistic regime reads

$$-\frac{\partial U_0(t,z)}{\partial t} = \max_{p_1}\left\{D_0(p_1 - c_0) - z\frac{\partial U_0}{\partial z}(V_0 - \alpha_0 p_1)\right\},$$

with the boundary condition $U_0(\tau, z(\tau)) = U_1(\tau, z(\tau)) = H(\tau)z$. Similarly to the identification of the value function in the competitive regime, $U_0(t, z)$ is conjectured to be of the form $U_0(t, z) = G(t)z$. Inserting the FOC $p_D = p_1 = 0.5(c_0 + V_0/\alpha_0 + \partial\ U_0/\partial z)$ into the HJB, gives a non-linear ODE which solution is $G(t)$,

$$G(t) = \frac{4H(\tau)\alpha_0 - (t-\tau)(V_0 - c_0\alpha_0)(V_0 - (H(\tau) + c_0)\alpha_0)}{\alpha_0(4 - (t-\tau)(V_0 - (H(\tau) + c_0)\alpha_0))}, \tag{30}$$

which decreases in t, implying that price skimming is optimal in the monopolistic regime.

A.2. Proof Proposition 2

The first-order conditions for (17) are

$$p_i = \frac{\gamma p_{3-i} + V_i + \frac{\partial W_i}{\partial z}\alpha_i + c_i(\gamma + \alpha_i)}{2(\gamma + \alpha_i)}. \tag{31}$$

Next, we insert the solution $\{p_1^*, p_2^*\}$ in (17). We conjecture player i's post-entry value function to be proportional to z such that $W_i(z) = A_i z$. The coefficients of the value functions solve the following algebraic equations for $i = \{1, 2\}$,

$$\omega A_i = \eta_{0i} + \eta_{1i}A_i + \eta_{2i}A_{1i}^2 + \eta_{3i}A_{3-i}, \tag{32}$$

where $\eta_{0i}, \eta_{1i}, \eta_{2i}, \eta_{i3}$ are functions of the model's parameters. Since the value functions are linear in z we have $\frac{\partial W_i}{\partial z} = A_i$ and $\frac{\partial W_{3-i}}{\partial z} = A_{3-i}$, which implies that the optimal pricing strategies are constant over time. Assuming that firms are symmetric, we note that there are two roots to the solutions of (32), but only the smallest one guarantees $\dot{z}(t) < 0$, i.e.

$$A_2 = A_1 = \frac{-y_2 - \sqrt{y_2^2 - 4y_1y_3}}{2y_1} > 0, \tag{33}$$

with $y_1 = \frac{\alpha_1^2(3\alpha_1+2\gamma)}{(2\alpha_1+\gamma)^2}, y_2 = -\frac{\left(5\alpha_1\gamma+4\alpha_1^2+2\gamma^2\right)(V_1-c_1\alpha_1)}{(2\alpha_1+\gamma)^2} - \omega_1$ and $y_3 = \frac{(\alpha_1+\gamma)(V_1-c_1\alpha_1)^2}{(2\alpha_1+\gamma)^2}$. We conduct sensitivity analyses with respect the model parameters to obtain the following results

$$\frac{\partial A_1^*}{\partial\gamma} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\gamma} = \frac{\alpha_1\left(c_1 - A_1 + (\gamma+2\alpha_1)\frac{\partial A_1}{\partial\gamma}\right) - V_1}{(\gamma+2\alpha_1^2)} < 0; \quad \frac{\partial A_1^*}{\partial\omega_1} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\omega_1} = \frac{\alpha_1\frac{\partial A_1}{\partial\omega_1}}{\gamma+2\alpha_1} < 0;$$

$$\frac{\partial A_1^*}{\partial c_1} > 0 \Rightarrow \frac{\partial p_1^*}{\partial c_1} = \frac{\gamma+\alpha_1\left(1+\frac{\partial A_1}{\partial c_1}\right)}{\gamma+2\alpha_1} > 0; \quad \frac{\partial A_1^*}{\partial V_1} > 0 \Rightarrow \frac{\partial p_1^*}{\partial V_1} = \frac{1+\alpha_1\frac{\partial A_1}{\partial V_1}}{\gamma+2\alpha_1} > 0;$$

$$\frac{\partial A_1^*}{\partial\alpha_1} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\alpha_1} = \frac{-\gamma c_1 - 2V_1 + \gamma A_1 + \alpha_1(\gamma+2\alpha_1) + \frac{\partial A_1}{\partial\alpha_1}}{(\gamma+2\alpha_1)^2} < 0.$$

A.3. Proof Proposition 3

The first-order condition of (22) with respect to p_0 yields

$$p_1^* = \frac{V_0 + \alpha_0(\partial W_0/\partial z_0 + c_0)}{2\alpha_0}. \tag{34}$$

We conjecture that the value function is proportional to z, such that $W_0(z) = A_0 z$. Consequently, the optimal pricing strategy in the monopolistic regime equals

$$p_1^* = \frac{V_0 + A_0\alpha_0 + c_0\alpha_0}{2\alpha_0}, \tag{35}$$

with

$$A_0 = \frac{-x_2 - \sqrt{x_2^2 - 4x_1x_3}}{2x_1} > 0, \tag{36}$$

with $x_1 = \frac{\alpha_0}{4}$, $x_2 = \frac{1}{2}(c_0\alpha_0 - V_0 - 2(\omega_0+\chi))$ and $x_3 = \frac{4A_1\chi\alpha_0 + (V_0 - c_0\alpha_0)^2}{4\alpha_0}$, which guarantees $\dot z(t) < 0$, as well and implies that $\partial p_1/\partial t = 0$. The sensitivity analyses of the pre-entry pricing strategy with respect to the pre- and post-entry[6] parameters are

$$\frac{\partial A_0}{\partial\gamma} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\gamma} < 0; \quad \frac{\partial A_0}{\partial\omega_0} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\omega_0} < 0;$$

$$\frac{\partial A_0}{\partial\omega_1} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\omega_1} < 0; \quad \frac{\partial A_0}{\partial V_0} > 0 \Rightarrow \frac{\partial p_1^*}{\partial V_0} > 0;$$

$$\frac{\partial A_0}{\partial V_1} > 0 \Rightarrow \frac{\partial p_1^*}{\partial V_1} > 0; \quad \frac{\partial A_0}{\partial c_0} > 0 \Rightarrow \frac{\partial p_1^*}{\partial c_0} > 0;$$

$$\frac{\partial A_0}{\partial c_1} > 0 \Rightarrow \frac{\partial p_1^*}{\partial c_1} > 0; \quad \frac{\partial A_0}{\partial\alpha_0} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\alpha_0} < 0;$$

$$\frac{\partial A_0}{\partial\alpha_1} < 0 \Rightarrow \frac{\partial p_1^*}{\partial\alpha_1} < 0.$$

The sensitivity of the pre-entry pricing strategy with respect to χ, i.e. $\frac{\partial A_0}{\partial\chi}$, reveals a dual role for χ. The sign of $\frac{\partial p_0}{\partial\chi}$ depends on the roots of the second-order polynomial $\Delta(A_1) = 4A_1\alpha_0\omega_0 - (V_0 - \alpha_0(A_1 + c_0))^2$, ρ_1 and ρ_2 equal to

$$\rho_1 = \frac{(-c_0\alpha_0 + V_0 + 2\omega_0) - \sqrt{\left(4\omega_0(V_0 - c_0\alpha_0) - (V_0 - c_0\alpha_0)^2 + 4\omega_0^2\right)}}{2\alpha_0},$$

$$\rho_2 = \frac{(-c_0\alpha_0 + V_0 + 2\omega_0) + \sqrt{\left(4\omega_0(V_0 - c_0\alpha_0) - (V_0 - c_0\alpha_0)^2 + 4\omega_0^2\right)}}{2\alpha_0^2}.$$

$\frac{\partial p_0}{\partial\chi} > 0$, for $\rho_1 < A_1 < \rho_2$, whereas if $A_1 < \rho_1$ or $A_1 > \rho_2$, then $\frac{\partial p_1}{\partial\chi} < 0$. However, we conjecture that $A_1 > \rho_2$ cannot happen since it would imply that $\dot z(t) < 0$, which is not possible economically. Hence, $\frac{\partial p_1}{\partial\chi} > 0$ if and only if the coefficient of the value function after the imitative entry is above ρ_1. Otherwise, $\frac{\partial p_1}{\partial\chi} < 0$.

A.4. Proof Proposition 4

When the date of the technological breakthrough, i.e. T, is known, the value function of the incumbent in the competitive regime is given by $U_1(t, z) = H(t)z$ for $t > \tau$, with $H(t)$ defined in (29). To characterize the optimal strategy and value function for the incumbent in the monopolistic regime, we note that $N_0(\tau, z) = U_1(\tau, z)$. Hence, the objective function of the incumbent reads

$$N_0(0, z(0)) = \max_{p_1} \mathbb{E}_\Gamma\left\{\int_0^\tau D_0(t)(p_1(t) - c_0)dt + U_1(\tau, z)\right\}, \tag{37}$$

s.t. (6). Since τ is the random variable governed by Γ, we recast the incumbent's problem as

$$N_0(t, z) = \max_{p_1} \int_0^T \chi e^{-\chi\tau}\left\{\int_0^\tau (D_0(s)(p_0(s) - c_0)ds + N_1(\tau, z(\tau)))\right\}d\tau. \tag{38}$$

Next, we integrate (38) by parts to obtain

$$N_0(0, z(0)) = \max_{p_1} \int_0^T e^{-\chi t}\{D_0(t)(p_1(t) - c_0) + \chi N_1(t, z)\}dt. \tag{39}$$

The corresponding HJB equation reads

$$-\frac{\partial N_0(t,z)}{\partial t} = \max_{p_1}\left\{D(t)(p_1(t) - c_0) + \chi N_1(t,z) - z\frac{\partial N_0(t,z)}{\partial z}(V_0 - \alpha_0 p_1)\right\}, \tag{40}$$

which yields the first-order condition

$$p_1(t) = \frac{V_0 + c_0\alpha_0 + \alpha_0\partial N_0/\partial z}{2\alpha_0}. \tag{41}$$

Inserting (41) in (40) and conjecturing that the value function is separable, i.e. $N_0(t, z) = E(t)z$, we obtain the following Riccati equation by the method of undetermined coefficients

$$\dot E(t) = -\frac{(V_0 - c_0\alpha_0)^2}{4\alpha_0} - H(t) + \left(\frac{V_0 - c_0\alpha_0}{2} + \chi\right)E(t) - \frac{\alpha_0}{4}E(t)^2. \tag{42}$$

To characterize the evolution over time of the incumbent's price, it suffices to determine the sign of $\dot E(t)$ since $\frac{\partial p_1^*}{\partial t} = \frac{1}{2}\dot E(t)$. To do so, we note that the equation of motion of $\dot E(t)$ is a second order polynomial in $E(t)$. We assume for economic reasons that $\forall \in [0, T]$, the condition $E(t) \geqslant 0$ is true, which implies that (i) $\dot E(t)$ is concave in $E(t)$ and (ii) the maximum is reached at $\overline{E}(t) = \frac{V_0 - \alpha_0 c_0 + 2\chi}{\alpha_0}$. Consequently, the roots of the polynomial, i.e. b_1 and b_2, are equal to

$$b_1 = \frac{2\chi + V_0 - c_0\alpha_0 - 2\sqrt{\chi(\chi + V_0 - \alpha_0(c_0 + H(t)))}}{\alpha_0} > 0,$$

$$b_2 = \frac{2\chi + V_0 - c_0\alpha_0 + 2\sqrt{\chi(\chi + V_0 - \alpha_0(c_0 + H(t)))}}{\alpha_0} > 0.$$

[6] Note that $\partial A_0/\partial A_1 > 0$, therefore we use the results in the proof of Proposition 2 to derive the sensitivity analyses of p_0 with respect to the post-entry parameters.

These results imply that if $E(t) < b_1$ or $E(t) > b_2(t)$, $\dot{E}(t)$ is negative. Conversely, if $b_1(t) \leqslant E(t) \leqslant b_2(t) \Rightarrow \dot{E}(t) > 0$. Hence there are two stable equilibria 0 and b_2. Moreover, we know that at $t = T$, $E(T) = 0$ and we shall look for a stable equilibrium that guarantees this boundary condition. The only solution that will ensure the stability of the equilibrium is to have $E(0) < b_1$. Indeed, if $E(0) > b_1$, $E(t)$ increases until $\overline{E}(t)$ is reached, at which time the sign of $\dot{z}(t)$ becomes positive. This means that the pool of unserved customers is increasing as t increases, which is obviously not possible for economic reasons. In addition, when $E(0) > b_1$, $E(t)$ converges toward b_2 which (i) leads to $\dot{z}(t) > 0$ and (ii) is different from 0. Therefore, the system has to start with $E(0) < b_1$ to ensure that $E(t)$ converges toward the only stable equilibrium that is economically possible, i.e. $E(0) < b_1$, which implies that $\dot{E}(t) < 0$ and consequently $\partial p_1^*/\partial t < 0$ in the monopoly regime.

A.5. Proof Proposition 5

Similar to previous propositions, we proceed backwards to characterize the pricing strategies in each regime. Consider first the competitive regime. The imitative entrant entered the market at τ. The date of the technological breakthrough is unknown, which implies that firms' optimization programs are

$$W_i(\tau, z) = \max_{p_i} \mathbb{E}_{\Omega_1}\left\{\int_\tau^T D_i(t)(p_i(t) - c_i)dt\right\}, \tag{43}$$

s.t. (7). The characterization of the optimal pricing strategies and value functions is identical to the proof of Proposition 2, i.e. firms choose constant prices in the competitive regime. Before the competitive entry takes place, the incumbent knows the date of the imitative entry but ignores the date of innovative entry. Therefore, her problem is

$$F_0(0, z(0)) = \max_{p_1} \mathbb{E}_{\Omega_0}\left\{\int_0^{\min\{\tau,T\}} D_0(t)(p_1(t) - c_0)dt + W_1(\tau, z)\cdot \mathbf{1}_{[\tau<T]}\right\}, \tag{44}$$

s.t. (6). Note that the difference between (44) and (8) is the information set conditioning the expectation operator, respectively Ω_0 only vs. both χ and Ω_0. The HJB equation for (44) reads

$$-\frac{\partial F_0(t,z)}{\partial t} = \max_{p_1}\left\{D_0(p_1 - c_0) - \omega_0 F_0(t,z) - z\frac{\partial F_0}{\partial z}z(V_0 - \alpha_0 p_1)\right\}, \tag{45}$$

with the boundary condition $F_0(\tau, z) = A_1 z$. The first order condition is

$$p_1(t) = \frac{V_0 + c_0\alpha_0 + \alpha_0 \partial F_0(t,z)/\partial z}{2\alpha_0}. \tag{46}$$

After inserting (46) in (45), the structure of the value function is conjectured to follow $F_0(t, z) = R_0(t)z$. As a consequence, we obtain the following differential equation by the method of undetermined coefficients:

$$\dot{R}_0(t) = \omega_0 R_0(t) - \frac{(V_0 - c_0\alpha_0 - \alpha_0 R_0(t))^2}{4\alpha_0}, \tag{47}$$

with the boundary condition $R_0(\tau) = A_1$. Similarly to the previous proposition, the dynamics of $R_0(t)$ is governed by a second-order polynomial. The only stable equilibrium that guarantees $\dot{z}(t) < 0$ is such that $R(t) < r_1$, where r_1 is the smallest root, which implies $\dot{R}_0(t) < 0 \Rightarrow \partial p_1/\partial t < 0$.

References

Aghion, P., & Howitt, P. (1998). *Endogeneous Growth Theory*. MIT Press.
Basar, T., & Haurie, A. (1984). Feedback equilibria in differential games with structural and modal uncertainties. *Advances in Large Scale Systems, 1*, 163–201.
Bensoussan, A., & Sethi, S. P. (2007). The machine maintenance and sale age model of Kamien and Schwartz revisited. *Management Science, 53*, 1964–1976.
Besanko, D., Dranove, D., Shanley, M., & Schaefer, S. (2009). *Economics of Strategy*. John Wiley and Sons.
Besanko, D., & Winston, W. L. (1990). Optimal price skimming by a monopolist facing rational consumers. *Management Science, 36*, 555–567.
Bourguignon, F., & Sethi, S. P. (1981). Dynamic optimal pricing and (possibly) advertising in the face of various kinds of potential entrants. *Journal of Economic Dynamics and Control, 3*, 119–140.
Bower, J. L., & Christensen, C. M. (1995). Disruptive technologies: catching the wave. *Harvard Business Review*, 43–53.
Chatterjee, R. (2009). Strategic pricing of new products and services. In V. Rao (Ed.), *Handbook of Pricing Research in Marketing*. Edward Elgar Publishing.
Chatterjee, R., & Eliashberg, J. (1990). The innovation diffusion process in a heterogeneous population: a micromodeling approach. *Management Science, 36*, 1057–1079.
Chen, Y.-M., & Jain, D. C. (1992). Dynamic monopoly pricing under a poisson-type uncertain demand. *The Journal of Business, 65*, 593–614.
Chenavaz, R. (2012). Dynamic pricing, product and process innovation. *European Journal of Operational Research, 222*, 553–557.
Chevalier-Roignant, B., Flath, C. M., Huchzermeier, A., & Trigeorgis, L. (2011). Strategic investment under uncertainty: a synthesis. *European Journal of Operational Research, 215*, 639–650.
Christensen, C. M. (1997). *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail*. Harvard Business School Press.
DeGiovanni, P. (2011). Quality improvement vs. advertising support: which strategy works better for a manufacturer. *European Journal of Operational Research, 208*, 119–130.
Deshmukh, S. D., & Chikte, S. D. (1976). Dynamic pricing with stochastic entry. *The Review of Economic Studies, 43*, 91–97.
Dockner, E., & Fruchter, G. (2004). Dynamic strategic pricing and speed of diffusion. *Journal of Optimization Theory and Applications, 123*, 331–348.
Dockner, E., & Gaunersdorfer, A. (1996). Strategic new product pricing when demand obey saturation effects. *European Journal of Operational Research, 90*, 589–598.
Dockner, E., & Jørgensen, S. (1988). Optimal pricing strategies for new products in dynamic oligopolies. *Marketing Science, 7*, 315–334.
Dockner, E., Jørgensen, S., Long, N., & Sorger, G. (2000). *Differential Games in Economics and Management Science*. Cambridge University Press.
Eliashberg, J., & Jeuland, A. P. (1986). The impact of competitive entry in a developing market upon dynamic pricing strategies. *Marketing Science, 5*, 20–36.
Erickson, G. M. (2012). Transfer pricing in a dynamic marketing–operations interface. *European Journal of Operational Research, 216*, 326–333.
Fruchter, G. E., & Messinger, P. R. (2003). Optimal management of fringe entry over time. *Journal of Economic Dynamics and Control, 28*, 445–466.
Gaskins, D. W. (1971). Dynamic limit pricing: optimal pricing under threat of entry. *Journal of Economic Theory, 3*, 306–322.
Geroski, P. A. (1999). *Early Warning of New Rivals*. Spring, pp. 107–116.
Hernandez-Mireless, C., Fok, D., Franses, P.H., 2012. Modeling the Triggers, Timing, and Speed of New Product Price Landings. Working Paper, University of Southern California.
Huang, J., Leng, M., & Liang, L. (2012). Recent developments in dynamic advertising research. *European Journal of Operational Research, 220*, 591–609.
Jørgensen, S., & Zaccour, G. (2004). *Differential Games in Marketing*. Kluwer Academic Publishers.
Kamien, M. I., & Schwartz, N. L. (1971). Limit pricing and uncertain entry. *Econometrica, 39*, 441–454.
Kamien, M. I., & Schwartz, N. L. (1983). *Market Structure and Innovation*. Cambridge University Press.
Kamrad, B., Lele, S. S., Siddique, A., & Thomas, R. J. (2005). Innovation diffusion uncertainty, advertising and pricing policies. *European Journal of Operational Research, 164*, 829–850.
Lambrecht, B., & Perraudin, W. (2003). Real options and preemption under incomplete information. *Journal of Economic Dynamics and Control, 27*, 619–643.
Moore, D. A., Oesch, J. M., & Zietsma, C. (2007). What competition? Myopic self-focus in market-entry decisions. *Organization Science, 18*, 440–454.
Pennings, E., & Sereno, L. (2011). Evaluating pharmaceutical R and D under technical and economic uncertainty. *European Journal of Operational Research, 212*, 374–385.
Rao, V. R., & Kartano, B. (2009). Pricing objectives and strategies: a cross-country survey. In V. R. Rao (Ed.), *Handbook of Pricing Research in Marketing*. Edward Elgar Publisher.
Robinson, B., & Lakhani, C. (1975). Dynamic price models for new-product planning. *Management Science, 21*, 1113–1122.